Threshold Aware Cosine Similarity Driven Box Projection and Weighted Fusion for Confidence Recovery in Single Drone Tracking

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Abstract*—Accurate UAV (Unmanned Aerial Vehicle) detection in aerial videos is often hindered by fluctuating confidence scores, especially under motion blur, scale variation, and other noise that may be present. To address this, we propose a Temporal Confidence Restoration Framework that exploits inter-frame motion information through a buffer-based cosine similarity mechanism. It is based on the fact that confidence drops likely occur when the drone is in motion due to effects of motion blur, but we can leverage this fact to restore and reinforce confidence scores based on motion trajectory because a UAV moving at a good speed cannot change directions abruptly. Our approach is used for a single drone, that first normalizes drone displacements across a temporal buffer to estimate consistent motion direction and magnitude. Using this motion cue, previous bounding boxes are projected into the current frame as near, medium, and far candidates, and also introduce a perpendicular jitter component based on cosine similarities to handle overlap. A Circular Weighted Boxes Fusion (WBF) is used on these projected boxes, with their weight derived as a function cosine similarity and average confidence values of the buffer boxes to get a more generalized detection box. Experimental evaluations on Drone Tracking video datasets with no false positive detections by the base model demonstrate a mean confidence increase of 2-3% over the videos, where roughly 55% of the time the system is triggered, and a reinforcement of confidence scores up to 4-5% is observed when the UAV is under appreciable motion. Hence our method improves temporal stability and confidence reliability and the natural reinforcement of detection score without requiring retraining or additional supervision. The codes and datasets are available at: https://github.com/BhavyaK31/UAV-CIRCLE-WBF

***Index Terms*—UAV Detection, Temporal Confidence Restoration, Cosine Similarity, Weighted Box Fusion, Motion Buffer, Drone Tracking**

I. Introduction

In recent years, the rapid growth of unmanned aerial vehicles (UAVs) has introduced significant challenges in airspace management and security monitoring. The growing number of unauthorized or uncontrolled drone flights puts public safety and restricted areas at risk [1]. Detecting and tracking such small UAVs is particularly difficult due to their low-altitude flight paths, relatively slow velocities, and compact sizes [2]. This reduces the effectiveness of traditional radar based and radio based detection systems [3], [4]. Hence, vision-based UAV detection has emerged as a practical and flexible solution [5]. However, under motion blur, low resolution, and variable illumination, confidence scores for detections decrease, but can be reconstructed based on motion of the drone, hence motivating this work.

Representative algorithms include Faster R-CNN and SSD (Single Shot Multi Box Detector, SSD) [6] based on two-stage detection mechanism and YOLO (You Only Look Once, YOLO) series based on one-stage detection mechanism.

Spatiotemporal UAV tracking methods, like [7] use systems for lightweight adaptive detection with multi-scale feature extraction, spatiotemporal motion modelling through Kalman-filter-based trajectory prediction and composite scoring based on detection confidence, appearance similarity and motion consistency and have impressive results, however for only single drone confidence regeneration simpler pipelines without the use of multi-scale feature extraction can be utilized to save time and computational power.

TCTrack: Temporal Contexts for Aerial Tracking [8] is another robust aerial tracking method that explicitly models temporal context by using transformers and uses these to build similarity features for better target localization. For cases where the base model ensures good spatial capture and fewer false positives, a simpler method along the same lines of using temporal geometric direction similarity for a drone moving with enough speed can be used, which saves time and computation.

In this work, we propose a simple and lightweight motion-aware temporal-consistency-driven drone tracking framework that leverages pipeline triggering for UAV speed over a threshold, box buffer displacement modeling, cosine similarity–based box projection and circular weighted box fusion (CWBF) to enhance target stability in aerial sequences. Unlike conventional frame-to-frame association approaches that depend solely on appearance or direct motion cues, our method maintains a temporal buffer of past detections and computes normalized displacement vectors across these frames to capture short-term motion dynamics. On the current frame t, we compute the cosine similarity between the most recent motion vector and the historical buffer trends to identify

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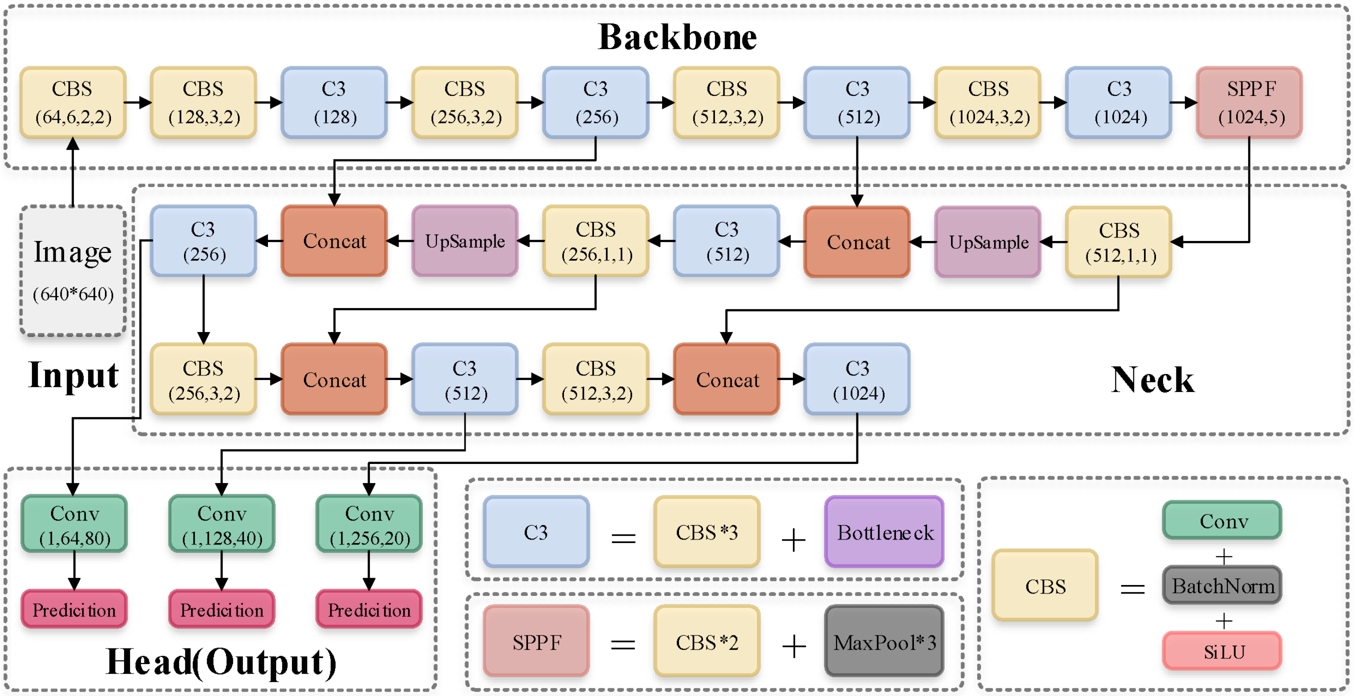


Fig. 1: Default Network Structure of YOLOv5 [9]

motion continuity and project near, medium, and far candidate boxes around the last known target location. To introduce better projections, a perpendicular jitter component proportional to the direction is introduced for the projected boxes. Finally, a circular weighted box fusion mechanism integrates the projected and detected boxes, assigning adaptive weights based on the base confidence score of the current box and average confidence score over the buffer. This fusion yields refined bounding boxes and confidence estimates that maintain temporal coherence without requiring deep feature embeddings or heavy temporal models. Consequently, our method provides a lightweight yet temporally and velocity aware tracking strategy, particularly effective in restoring confidence scores and getting better generalized scores and boxes when the base model can track effectively but lose confidence score due to motion blur, intensity changes and other noise effects.

II. Related Work

A. *YOLO Detection Models*

YOLO [10] based detectors have become the foundation for many modern tracking pipelines due to their real-time performance and strong generalization across object scales. Variants such as YOLOv4 [11], YOLOv7 [12], and YOLOv10 [13] provide precise bounding boxes that serve as reliable anchors for motion estimation and temporal association. In tracking, YOLO outputs are often combined with motion models or re-identification features to maintain object continuity across frames. Our approach similarly uses YOLOv5m [14](see Fig.1), which comprises of 4 parts: input, Darknet-53 [15] backbone network, PANet [16] neck network, and output. We use YoloV5m detections as initial observations, but enhance them with temporal displacement buffers, cosine similarity–based weighting, and weighted box fusion (WBF) to improve stability in motion scenarios.

B. *Frame Displacement Based Tracking*

Tracking objects across video frames relies heavily on modeling motion consistency and temporal relationships. Classical approaches estimate frame-to-frame displacements using optical flow or Kalman filters, where object position in the current frame is predicted from its previous trajectory. Such motion-based strategies exploit the assumption that object displacement between consecutive frames is smooth and locally coherent. When combined with modern detectors such as YOLO, these methods help bridge detection gaps and maintain identity continuity in challenging scenes. Recent works extend this by learning displacement embeddings or integrating frame differencing into deep networks to enhance robustness against occlusion and motion blur. For instance, SiamMask [17] and TCTrack [8] incorporate temporal context or displacement features to refine tracking decisions. These techniques align closely with our approach, which maintains a buffer of previous bounding boxes, models cosine similarity across displacement vectors, and refines predictions through weighted box fusion (WBF) to ensure stable, confidence-aware tracking.

C. *Weighted Boxes Fusion*

The WBF algorithm is a new method proposed by Roman to fuse the results of different detection models [18]. Unlike NMS [19] through the threshold to remove predicted boxes or Soft NMS [20] to decay box scores, WBF uses the confidence scores of all predicted boundary boxes to construct the average

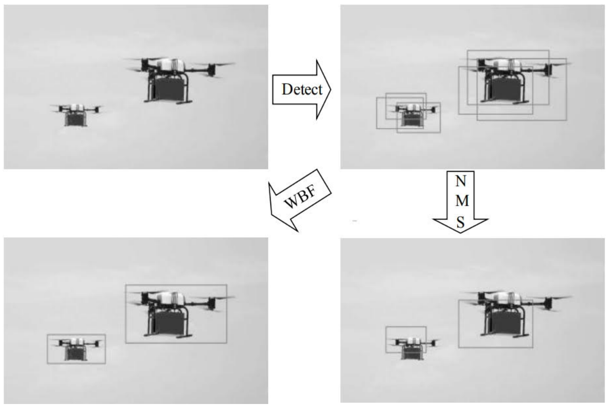


Fig. 2: Comparison of WBF ans NMS [21]

box(see Fig.2). Therefore, this fusion method can significantly improve the accuracy of the prediction results under the condition that the quality of the predicted box is good. With good ensemble characteristics, this method has achieved good results in Open Images, COCO object detection challenges and Kaggle competitions. In our work we convert boxes to circles and then use same algorithm as WBF, then convert back to boxes.

III. Model Design

This section mainly introduces the overall model architecture of the additional pipeline and explains the principle and the intuition involved in it.

A. Movement Threshold

When the speed of the drone is low, we cannot assume a direction that the drone takes. However, detecting a stationary target is doable with more confidence as compared to a moving target, so here we can rely on YOLO's confidence score, and empty the buffers for previous frames as they will not give reliable projection of boxes.

$$B_t = \begin{cases} B_{\text{YOLO}}, & \text{if } \Delta_m < \tau * d_{\text{YOLO}} \\ B_{\text{pipeline}}, & \text{if } \Delta_m \geq \tau * d_{\text{YOLO}} \end{cases}$$

B. Historical Displacement Computation

A buffer-based displacement computation captures both the magnitude and direction of the historical motion. The averaged normalized vector $\bar{d}$ represents the motion trend, while $\hat{d}_t$ provides the instantaneous motion direction. These are later compared through cosine similarity to estimate motion consistency and enhance temporal fusion.
To maintain temporal consistency of drone detections, a buffer of YOLO bounding boxes from the previous k frames is maintained:

$$\mathcal{B} = \{B_{t-k}, B_{t-k+1}, \ldots, B_{t-1}\}.$$

Each bounding box B_i is represented by its center coordinate:

$$C_i = (x_i, y_i).$$

The historical displacement vectors between consecutive frames are computed as:

$$\Delta C_i = C_i - C_{i-1} = (x_i - x_{i-1},\ y_i - y_{i-1}), i = t-k+1 \ldots t-1$$

The corresponding displacement magnitude is given by:

$$m_i = \|\Delta C_i\|_2 = \sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2}.$$

Each displacement is then normalized to obtain the direction:

$$\hat{d}_i = \frac{\Delta C_i}{m_i + \varepsilon},$$

where $\varepsilon = 10^{-9}$ prevents division by zero for negligible motion.

The average normalized direction across the history buffer is calculated as:

$$\bar{d} = \frac{1}{N} \sum_{i=1}^{N} \hat{d}_i, \quad N = |\mathcal{B}| - 1,$$

and re-normalized to ensure unit length:

$$\bar{d} \leftarrow \frac{\bar{d}}{\|\bar{d}\|_2 + \varepsilon}.$$

In parallel, the set of historical displacement magnitudes is stored for later use in motion scaling:

$$M = \{m_1, m_2, \ldots, m_N\}.$$

For the current frame, the displacement vector between frames $t-1$ and t is computed as:

$$\Delta C_t = (dx, dy), \quad d_t = \sqrt{dx^2 + dy^2},$$

and its normalized form is:

$$\hat{d}_t = \frac{(dx, dy)}{d_t + \varepsilon}.$$

If insufficient history is available ($N = 0$), the current normalized displacement $\hat{d}_t$ is directly used as a fallback mean direction.

C. Closeness and Scale Estimation

Compute closeness from cosine similarity, use mean and standard deviation of previous displacement magnitudes to control degree of perpendicular jitter. We also calculate average area ratio along the buffer that can be used along with the nearer, medium and farther projections later.

a) Closeness Metric: Let $\mathbf{d}_{\text{avg}}$ denote the normalized average displacement vector from the buffer, and $\mathbf{d}_{\text{cur}}$ denote the current normalized displacement. The cosine similarity between these vectors quantifies directional consistency:

$$\text{cos\_sim} = \mathbf{d}_{\text{avg}} \cdot \mathbf{d}_{\text{cur}} \tag{1}$$

which lies in the range $[-1, 1]$. To map this to a confidence-like metric, we define

$$C = \frac{1 + \text{cos\_sim}}{2}, \quad C \in [0, 1] \tag{2}$$

where C ("closeness") represents the similarity between the recent motion direction and the current one.

b) Magnitude Variation: Let $\{m_i\}$ represent magnitudes of past displacement vectors. We compute their normalized standard deviation to characterize motion stability:

$$V_m = \begin{cases} \dfrac{\sigma(m_i)}{\mu(m_i)+\epsilon}, & \text{if } N \geq 2 \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where ϵ is a small constant to prevent division by zero. Lower V_m indicates consistent motion, while higher V_m corresponds to erratic movement.

c) Size Scaling: The current YOLO bounding box $b = [x_1, y_1, x_2, y_2]$ provides a characteristic object scale:

$$S = \sqrt{(x_2 - x_1)(y_2 - y_1)} \tag{4}$$

which is used to normalize motion-based perturbations.

d) Directional Jitter Control: To simulate realistic spatial perturbations, a perpendicular unit vector $\mathbf{p}$ is obtained as

$$\mathbf{p} = \frac{[-d_y, d_x]}{\| [-d_y, d_x] \| + \epsilon} \tag{5}$$

where $\mathbf{d} = [d_x, d_y]$ is the current normalized displacement. The jitter magnitude σ is adaptively determined as

$$\sigma = \sigma_0 \, S \, (1 + V_m)(1 - 0.5C) \tag{6}$$

where σ_0 is a tunable base coefficient. Thus, σ decreases for stable and directionally consistent motion.

e) Scale Adjustment: To adapt to zoom effects or drone depth variations, we estimate the mean area ratio across the recent frame buffer:

$$R_a = \text{clip}\left(\frac{1}{N-1} \sum_{i=2}^{N} \frac{A_i}{A_{i-1}}, \, 0.85, \, 1.15 \right) \tag{7}$$

where A_i denotes the area of the i-th bounding box. This ratio constrains scale variations to a realistic range.

f) Closeness-based Candidate Scaling: Depending on C, we define three candidate spatial scales:

$$\begin{aligned} \beta_{\text{near}} &= 1 - 0.5C, \\ \beta_{\text{mid}} &= 1.0, \\ \beta_{\text{far}} &= 1 + 0.5(1 - C) \end{aligned} \tag{8}$$

indicating progressively wider search regions as confidence decreases.

D. Candidate Box Generation and Spatial Sampling

After computing motion consistency and scale factors, we generate a set of candidate bounding boxes to account for possible drone movement between consecutive frames. This step allows the tracker to remain robust even when the motion is uncertain or the detection temporarily drops.

a) Predicted Centers: Let (x_c^{t-1}, y_c^{t-1}) denote the center of the bounding box in the previous frame, and (d_x, d_y) the normalized displacement vector scaled by its recent magnitude. For each candidate type $k \in \{\text{near}, \text{mid}, \text{far}\}$, we compute a predicted center:

$$(x_c^{(k)}, y_c^{(k)}) = (x_c^{t-1}, y_c^{t-1}) + \beta_k (d_x, d_y) \tag{9}$$

where β_k scales how far along the predicted motion direction we extrapolate the next position. A smaller β_k corresponds to a conservative (near) estimate, while a larger β_k explores farther displacements.

b) Perpendicular Jitter: To simulate realistic uncertainty and small lateral deviations, we add a perpendicular jitter vector to each candidate:

$$(x_c^{(k)}, y_c^{(k)}) \leftarrow (x_c^{(k)}, y_c^{(k)}) + s_k \, \mathbf{p} \, \sigma \tag{10}$$

where $\mathbf{p}$ is the unit vector perpendicular to motion, σ is the adaptive jitter magnitude from the previous section, and $s_k \in \{-1, 0, +1\}$ alternates the direction of jitter for variety. This ensures that even if the motion direction estimate is imperfect, nearby spatial hypotheses remain covered.

c) Candidate Box Dimensions: For each predicted center, we define the corresponding box width and height as:

$$(w^{(k)}, h^{(k)}) = (\max(2, w_{\text{YOLO}} \cdot \lambda_k), \max(2, h_{\text{YOLO}} \cdot \lambda_k)) \tag{11}$$

where $w_{\text{YOLO}}, h_{\text{YOLO}}$ are the YOLO-predicted box dimensions and λ_k are scale multipliers linked to each β_k. This allows the candidates to account for potential zoom or scale changes.

d) Bounding Box Conversion: Each box is then converted from its center-based representation to the standard corner form:

$$\begin{aligned} (x_1^{(k)}, y_1^{(k)}, x_2^{(k)}, y_2^{(k)}) = (&x_c^{(k)} - \tfrac{w^{(k)}}{2}, \, y_c^{(k)} - \tfrac{h^{(k)}}{2}, \\ &x_c^{(k)} + \tfrac{w^{(k)}}{2}, \, y_c^{(k)} + \tfrac{h^{(k)}}{2}) \end{aligned} \tag{12}$$

and clipped to remain within frame boundaries.

e) Final Candidate Set:: The resulting set of boxes is

$$\mathcal{B} = \{b_{\text{YOLO}}\} \cup \{b^{(\text{near})}, b^{(\text{mid})}, b^{(\text{far})}\} \tag{13}$$

where b_{YOLO} is the raw detector output, and the remaining terms are motion-projected candidates. Together, they form a compact yet diverse hypothesis pool for subsequent selection or weighted fusion, enabling smoother and more consistent tracking even under noisy conditions.

E. Circular Weighted Boxes Fusion (CWBF)

To address the limitations of traditional rectangular fusion technique for UAVs, we propose a *Circular Weighted Boxes Fusion (CWBF)* approach. Unlike standard Weighted Boxes Fusion (WBF) which averages overlapping rectangles, CWBF models each detection as a circle, offering rotational symmetry and smoother aggregation under uncertain motion.

a) Circle Representation: Each bounding box $b = [x_1, y_1, x_2, y_2]$ is first transformed into an equivalent circular form:

$$c = (x_c, y_c, r) \tag{14}$$

where (x_c, y_c) is the box center and r is the diagonal-based radius given by:

$$x_c = \frac{x_1 + x_2}{2}, \quad y_c = \frac{y_1 + y_2}{2}, \\ r = \sqrt{\left(\frac{x_2 - x_1}{2}\right)^2 + \left(\frac{y_2 - y_1}{2}\right)^2} \tag{15}$$

This transformation allows for a more uniform spatial representation, reducing bias toward box aspect ratios and better modeling drone motion blur or rotational movement.

b) Circular Intersection over Union.: To measure overlap between two detections $c_i = (x_i, y_i, r_i)$ and $c_j = (x_j, y_j, r_j)$, the circular IoU is computed analytically:

$$\text{IoU}(c_i, c_j) = \frac{A_{\text{inter}}(c_i, c_j)}{A_{\text{union}}(c_i, c_j)} \tag{16}$$

where the intersection area A_{inter} is obtained from the geometric overlap of two circles separated by distance $d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$:

$$A_{\text{inter}} = r_i^2 \cos^{-1}\left(\frac{d_{ij}^2 + r_i^2 - r_j^2}{2d_{ij}r_i}\right) + r_j^2 \cos^{-1}\left(\frac{d_{ij}^2 + r_j^2 - r_i^2}{2d_{ij}r_j}\right) \\ - \frac{1}{2}\sqrt{(-d_{ij} + r_i + r_j)(d_{ij} + r_i - r_j)(d_{ij} - r_i + r_j)(d_{ij} + r_i + r_j)} \tag{17}$$

and the union area is defined as:

$$A_{\text{union}} = \pi(r_i^2 + r_j^2) - A_{\text{inter}} \tag{18}$$

This measure ensures consistent overlap evaluation even when drones change orientation or camera angle.

c) Weighted Fusion Process: Given multiple detection sets $\mathcal{D}_n = \{(b_k^{(n)}, s_k^{(n)}, \ell_k^{(n)})\}$ from different detectors or frames, each box is converted into its circular form and filtered by score threshold τ_s:

$$\mathcal{C} = \{c_k^{(n)} : s_k^{(n)} > \tau_s\} \tag{19}$$

For each label ℓ, we cluster overlapping circles using the IoU threshold τ_{iou} and compute their weighted centroid:

$$x_f = \frac{\sum_i w_i s_i x_i}{\sum_i w_i s_i}, \quad y_f = \frac{\sum_i w_i s_i y_i}{\sum_i w_i s_i} \tag{20}$$

$$A_f = \frac{\sum_i w_i s_i A_i}{\sum_i w_i s_i}, \quad r_f = \sqrt{\frac{A_f}{\pi}} \tag{21}$$

where w_i denotes detector reliability weight, s_i is confidence, and $A_i = \pi r_i^2$ is the area of each circle. The fused confidence is defined as:

$$s_f = \frac{\sum_i w_i s_i^2}{\sum_i w_i s_i} \tag{22}$$

Finally, the fused circle (x_f, y_f, r_f) is converted back to the rectangular form $[x_1, y_1, x_2, y_2]$ for evaluation or visualization.

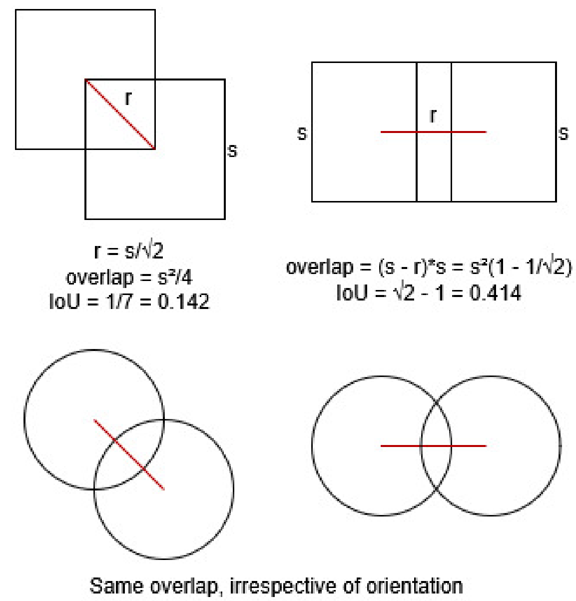


Fig. 3: Example of how CBWF performs better intuitively.

d) Intuition and Advantages: Conceptually, CWBF operates like WBF but smooths fusion spatially by enforcing circular isotropy. This helps stabilize bounding under drone jitter, partial occlusion, or non-uniform aspect ratios. The use of radius-based averaging instead of corner-based blending prevents elongated artifacts and better captures motion uncertainty in radial form. In practice, CWBF improves temporal consistency and robustness while preserving the interpretability and simplicity of the WBF framework.

F. Circular Weighted Boxes Fusion (CWBF)

Once all candidate boxes are generated, they are normalized and prepared for the Weighted Box Fusion (WBF) process. The goal is to merge overlapping candidate boxes into a single, motion-consistent detection. Let each candidate box be represented as $\mathbf{b}_i = [x_{1i}, y_{1i}, x_{2i}, y_{2i}]$ and normalized by the image width W and height H as follows:

$$\tilde{\mathbf{b}}_i = \frac{\mathbf{b}_i}{[W, H, W, H]}. \tag{23}$$

Next, the motion consistency of each candidate is computed based on the cosine similarity between the average historical displacement $\bar{\mathbf{d}}$ and the candidate's instantaneous displacement vector $\mathbf{v}_i = (v_{xi}, v_{yi})$ from the previous center:

$$\text{MC}_i = \max\big(0, \min(1, \cos(\theta_i))\big), \\ \text{where} \;\; \cos(\theta_i) = \frac{\bar{\mathbf{d}} \cdot \mathbf{v}_i}{\|\bar{\mathbf{d}}\| \, \|\mathbf{v}_i\|}. \tag{24}$$

Each candidate's fusion weight w_i is derived from the YOLO confidence c_y and the motion consistency factor MC_i. The historical mean of previous confidences, denoted $\bar{c}_{\text{prev}}$, is also incorporated to provide temporal stability:

$$w_i = c_y\Big(\bar{c}_{\text{prev}} + (1-\bar{c}_{\text{prev}})\,\text{MC}_i\Big), \quad \bar{c}_{\text{prev}} = \frac{1}{N}\sum_{j=1}^{N} c_{\text{prev},j}. \tag{25}$$

This ensures that boxes consistent with the object's motion direction receive greater weight, while abrupt or inconsistent displacements are down-weighted. Finally, all normalized boxes $\tilde{\mathbf{b}}_i$ and their corresponding weights w_i are passed to the WBF algorithm:

$$\{\mathbf{b}_f, c_f\} = \text{WBF}\big(\{\tilde{\mathbf{b}}_i\}, \{w_i\}\big), \tag{26}$$

where $\mathbf{b}_f$ and c_f denote the final fused bounding box and confidence, respectively.

If no fusion is possible (e.g., numerical instability or missing overlap), the algorithm robustly falls back to the YOLO-predicted box and confidence.

IV. Experimental Results

A. Dataset Preparation

1) Base YOLO: For the Yolov5m model, we use the weights given in [22]. These weights were obtained by training on two datasets, the Sky Background Dataset and the Complex Background Dataset. The Sky Background Dataset was obtained by merging three different datasets [23] [24] [25], using only the images with sky backgrounds. The Complex Background Dataset was made by merging 7 different datasets including [23] [24] [25] [26] [27] [28], using only the images with complex backgrounds, and adding images of birds flying to the dataset. The datasets were compiled in this paper [29]. The results from the paper on different tasks are shown below-

TABLE I: Yolov5m Performance

	mAP_{50}	$mAP_{50\text{-}95}$
Simple Background Dataset	87.7	71.7
Complex Background Dataset	94.6	59

2) Inference Videos: For video based inference where the method in this paper has been used for motion based confidence recovery, we take videos from [30], apply editing to remove any false positive detection to show the recovery performance when the base model is able to detect the single drone correctly. We also apply noising effects like motion blur, brightenss variations, etc, to get 97 clean videos and 67 noisy videos.

B. Experimental Setup

All experiments were conducted in python, and OpenCV was used for frame processing, and augmenting noise into the videos for the noisy videos. Each video was analyzed frame by frame, and the fusion method was triggered only when inter frame displacement of the YOLO bounding box exceeded the displacement threshold. So the method triggering and usage is deterministic and not based on any optimization algorithms.

For every video, we recorded:

(i) the number of frames where the fused confidence improved,

(ii) frames where the fused and YOLO confidences were equal

(iii) average confidence gain when fusion was active,

(iv) overall confidence gain across all frames,

(v) mean YOLO confidence

(vi) mean fused confidence, and

(vii) activation ratio, defined as the percentage of frames where the fusion module was triggered.

These metrics were computed for all clean and noise-augmented videos. Complete implementation details and code are available in the GitHub repository.

C. Evaluation Metrics

Because the dataset contains no false positives and the objective is to enhance the stability of frame-level confidence under drone motion, metrics such as mAP and IoU are not used. Instead, confidence change before and after applying the proposed conditional fusion module is used. The following metrics were computed for every video.

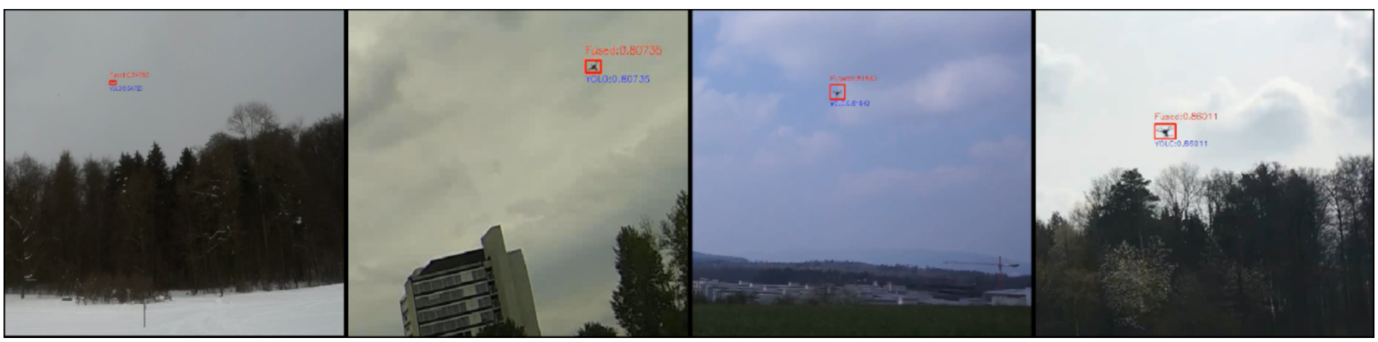
1) Confidence and Movement based Metrics:

- **Frames_Fused_Better:** Number of frames in which the fused confidence exceeded the original YOLO confidence. This method is mostly when drone movement exceeds displacement threshold and fusion method can be used beneficially, based on movement inertia.
- **Frames_Fused_Equal:** Number of frames where the fused and YOLO confidences were identical, mostly used when drone movement does not exceed displacement threshold and no movement based confidence losses occur.
- **Avg Accuracy Increase (WBF Only):** Mean increase in confidence over only those frames where fusion was activated. This isolates the effect of the fusion mechanism itself.
- **Avg Accuracy Increase (Overall):** Mean confidence gain computed over all frames in the video, representing the global effect of the proposed method across the sequence.
- **Avg YOLO Confidence:** Baseline detection confidence averaged across all frames, which acts as reference for assessing improvements.
- **Avg Fused Confidence:** Average detection confidence after applying the conditional fusion, which can be compared with average YOLO confidence.
- **Activation Ratio:** Ratio of frames where fused method was activated and all the detected frames. This ratio tells us percentage of times the fused method was used.

2) Justification for Metric Choice: Since YOLO does not do any false detections on the video dataset, we use the confidence recovery during appreciable motion as the metric, as per our objective. Evaluating confidence improvements across clean and noise-augmented videos enables assessment of both robustness and motion based consistency. The Activation Ratio provides information on how frequently the proposed method is used. Together, these metrics capture the operational and performance characteristics of the proposed method.

(a) Screenshots of frames from clear videos where wbf showed improvement

(b) Screenshots of frames from noisy videos where wbf showed improvement



(c) Screenshots of frames from clear videos where wbf did not show improvement

(d) Screenshots of frames from noisy videos where wbf did not show improvement

Fig. 4: Screenshots of frames from various clear and noisy videos.

D. Evaluation Results

This section reports the empirical results of the proposed fusion method in clean and noisy video sets. For both conditions, we present inferences from statistics, observations on confidence improvements, activation, and robustness.

1) Clean Dataset: Table II summarizes the descriptive statistics under clean conditions on 97 videos.

TABLE II: Descriptive Statistics for Clean Dataset

	Mean	Median	Std	Min	Max
Frames Used	227.38	217.00	133.74	3	564
Frames Not Used	256.58	155.00	277.27	1	1491
Gain (Active)	0.04576	0.04065	0.02589	1.65e-05	0.13624
Gain (Overall)	0.02324	0.01935	0.01668	4.89e-07	0.07509
YOLO Confidence	0.77501	0.77506	0.08190	0.53303	0.91066
Fused Confidence	0.79825	0.80918	0.08193	0.55397	0.91211
Activation Ratio	0.55550	0.52196	0.29643	0.00600	0.99679

The clean set shows:

- A mean confidence gain of **4.5%** on activated frames and **2.3%** overall.
- An activation ratio of **0.55**, indicating that WBF is triggered in over half the frames.
- Method confidence consistently exceeds YOLO baseline confidence by **2.3%**.

2) Noisy Dataset: Table III summarizes the statistics for noisy sequences on 67 videos.

TABLE III: Descriptive Statistics for Noisy Dataset

	Mean	Median	Std	Min	Max
Frames Used	170.87	141.00	123.23	11	630
Frames Not Used	199.93	107.00	235.76	2	1358
Gain (Active)	0.04600	0.04491	0.01861	0.01195	0.11536
Gain (Overall)	0.02329	0.02366	0.01356	0.00143	0.06484
YOLO Confidence	0.73262	0.76528	0.13303	0.40846	0.91277
Fused Confidence	0.75591	0.79420	0.13234	0.41397	0.92497
Activation Ratio	0.56008	0.54466	0.29971	0.01239	0.98438

The noisy set reveals:

- Confidence gains remain stable: **4.6%** (active) and **2.3%** (overall)—almost identical to clean data.
- YOLO baseline confidence drops significantly (from 0.77 to 0.73), but fused confidence maintains a similar advantage of **2.3%**.
- Activation ratio remains high (**0.56**), showing the method is noise-insensitive.

3) Result Inferences across both datasets:

- Mean and median confidence scores consistently increase after fusion.
- The improvement is stable (**2–3%**) in both datasets. This shows that as long as underlying detection is correct, the confidence recovery is robust under noise and only movement based.
- In the frames where drone movement is appreciable, the improvement is stable (**4-5%**) in both datasets. This is due to movement based recovery, as well as reinforcement of confidence that may have been lost due to motion shenanigans, as was observed in the video detections.
- Mean and median confidence scores differ by (**4-5%**) in clean and noisy videos. This shows that noise only affects base model confidence detection.
- Standard deviations of confidence based on YOLO and fused method do not differ that much.

V. Conclusion

This work presented a motion-triggered fusion mechanism designed for confidence recovery and reinforcement in single UAV detection. By using directional displacement cues and selectively applying fusion only when significant motion is detected, the proposed method is a lighweight addition to the underlying detector architecture. Experiments conducted on 150+ videos demonstrated that the method achieves an average confidence improvement of approximately 4-5% on frames where fusion is activated, and around 2% improvement over entire sequences. The activation ratio remained stable at roughly 55% across all videos, indicating that the selective activation mechanism only recovers and reinforces confidence score based on motion and not noise. Noise only effects the underlying model's detection capability. Furthermore, the fused confidence consistently exceeded the baseline YOLO confidence in both clean and degraded conditions, which is to be expected as the fusion does a circular weighted boxes fusion of projected boxes, for a better generalized drone box and confidence scores.

Future work could include the extension of this methods for multi-drone environments, where we could run this method for all the detected drones, and assign the drone IDs based on detected movement and features. Another simpler improvement could be the suppression of confidence scores of wrong detections, based on abrupt cosine similarity and spikes of confidence scores detected from the fusion method. This is based on our observation, that the fusion method shoots up the confidence scores by a lot when the sequence of detection boxes are unnatural and clearly do not follow an inertia based drone trajectory. This extension could provide some improvements in traditional metrics like mAP and IOU based on video datasets with ground truth boxes, and compare with similar architectures based on these traditional metrics.

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